Use the `head` command on your three files again. This time, describe at least one potential problem with the data you see. Consider issues with missing values and bad data.

For the `bus` dataframe, everything looks good, except that there is misformated data such as -9999 for longitude and latitude, as well as phone number -9999. These are probably missing values that were used as placeholders in the `.csv` files.

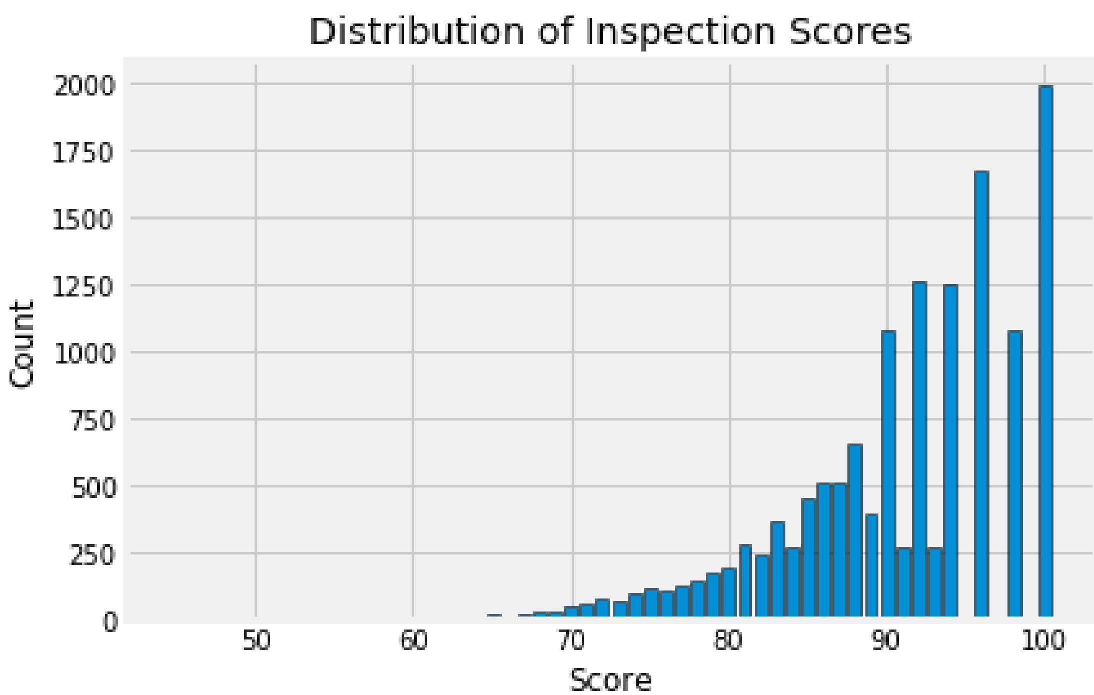
For the `ins` dataframe, everyhting looks good except for the scores, where some of them have -1 as a score, instead of 0 to 100. This is also probably missing data.

And finally, for the `vio` dataframe, we have good data, it seems like most or all of them are properly formatted or described well!

0.1 Question 5a

Let's look at the distribution of inspection scores. As we saw before when we called head on this data frame, inspection scores appear to be integer values. The discreteness of this variable means that we can use a bar plot to visualize the distribution of the inspection score. Make a bar plot of the counts of the number of inspections receiving each score.

It should look like the image below. It does not need to look exactly the same (e.g., no grid), but make sure that all labels and axes are correct.



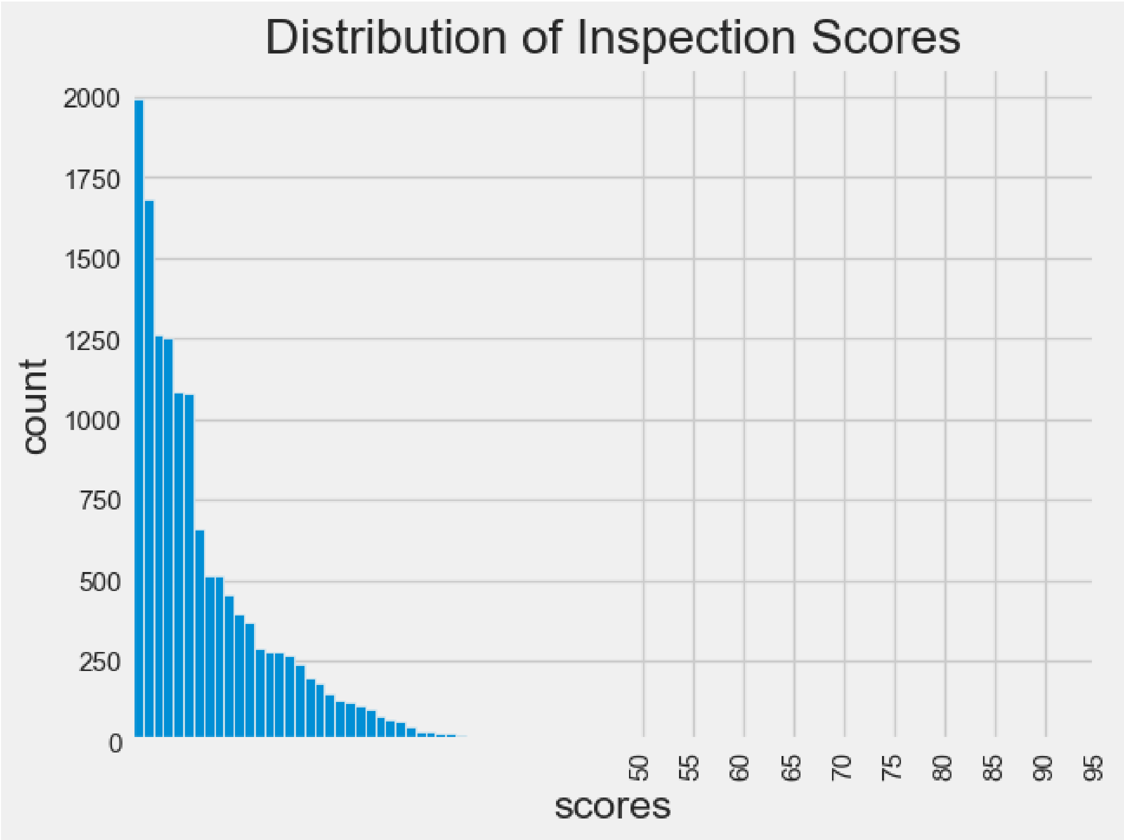
You might find this matplotlib.pyplot tutorial useful. Key syntax that you'll need:

```
plt.bar
plt.xlabel
plt.ylabel
plt.title
```

Note: If you want to use another plotting library for your plots (e.g. plotly, sns) you are welcome to use that library instead so long as it works on DataHub. If you use seaborn sns.countplot(), you may need to manually set what to display on xticks.

```
In [71]: score_counts = ins_named
         score_counts = score_counts[score_counts['score'] != -1]
         score_counts = score_counts['score'].value_counts()
         score_counts.plot(kind='bar', xticks=(np.arange(50, 100, 5)), title='Distribution of Inspectio
```

```
Out[71]: <AxesSubplot:title={'center':'Distribution of Inspection Scores'}, xlabel='scores', ylabel='co
```

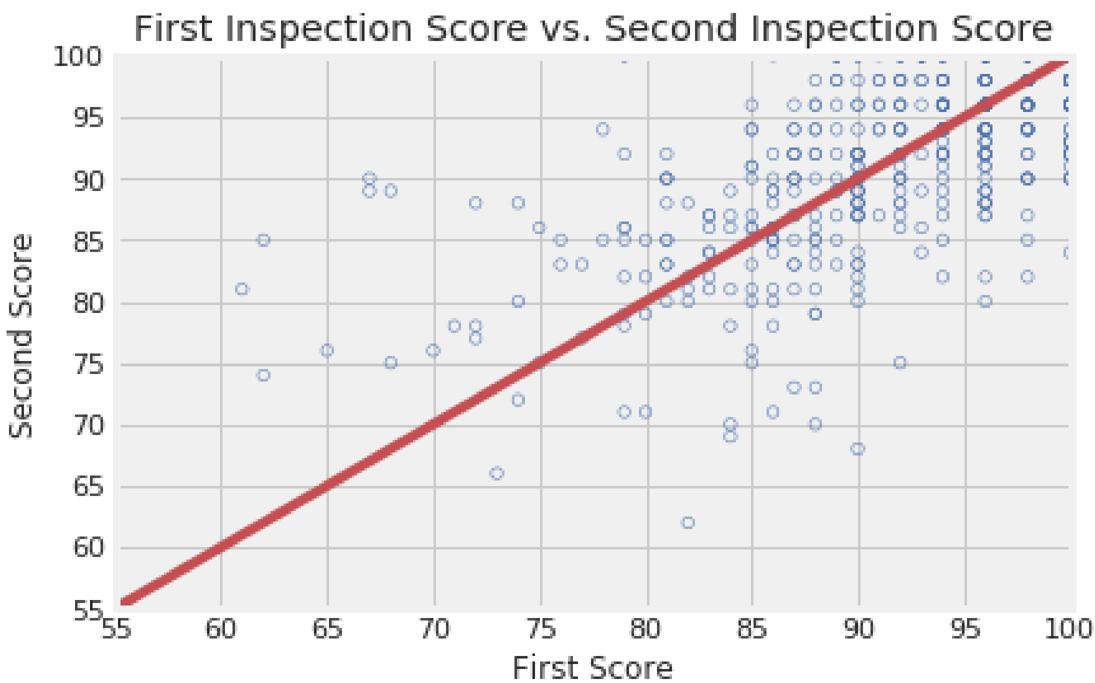


0.1.1 Question 5b

Describe the qualities of the distribution of the inspections scores based on your bar plot. Consider the mode(s), symmetry, tails, gaps, and anomalous values. Are there any unusual features of this distribution? What do your observations imply about the scores?

The scores are heavily on the upper half, and most of the scores are greater than 50. Thus, the distribution is skewed to the left. Inspection scores are usually high, implying that anything lower doesn't really get inspected, or gets thrown out of business for poor cleaniness. Also, every restaurant should be striving for a perfect inspection score to be kept in business, so it makes sense that the scores are near the higher side.

Now, create your scatter plot in the cell below. It does not need to look exactly the same (e.g., no grid) as the sample below, but make sure that all labels, axes and data itself are correct.



Key pieces of syntax you'll need:

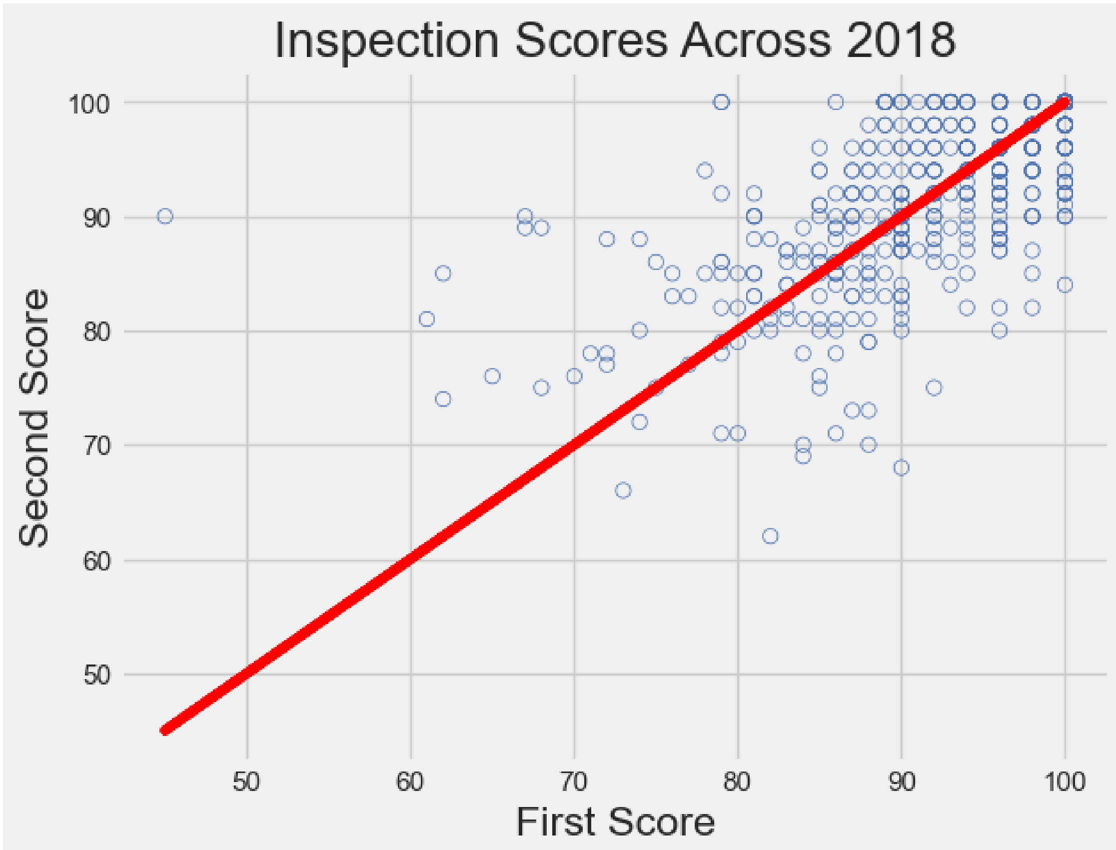
`plt.scatter` plots a set of points. Use `facecolors='none'` and `edgecolors='b'` to make circle markers with blue borders.

`plt.plot` for the reference line.

`plt.xlabel`, `plt.ylabel`, `plt.axis`, and `plt.title`.

Hint: You may find it convenient to use the `zip()` function to unzip scores in the list.

```
In [77]: x, y = list(zip(*scores_pairs_by_business['score_pair'].to_list()))
         plt.scatter(x, y, facecolors='none', edgecolors='b')
         plt.plot(x, x, color='red')
         plt.xlabel("First Score")
         plt.ylabel("Second Score")
         plt.title("Inspection Scores Across 2018");
```


Inspection Scores Across 2018
Second Score
First Score
100
90
80
70
60
50
50
60
70
80
90
100

0.1.2 Question 6c

If restaurants' scores tend to improve from the first to the second inspection, what do you expect to see in the scatter plot that you made in question 7c? What do you oberve from the plot? Are your observations consistent with your expectations?

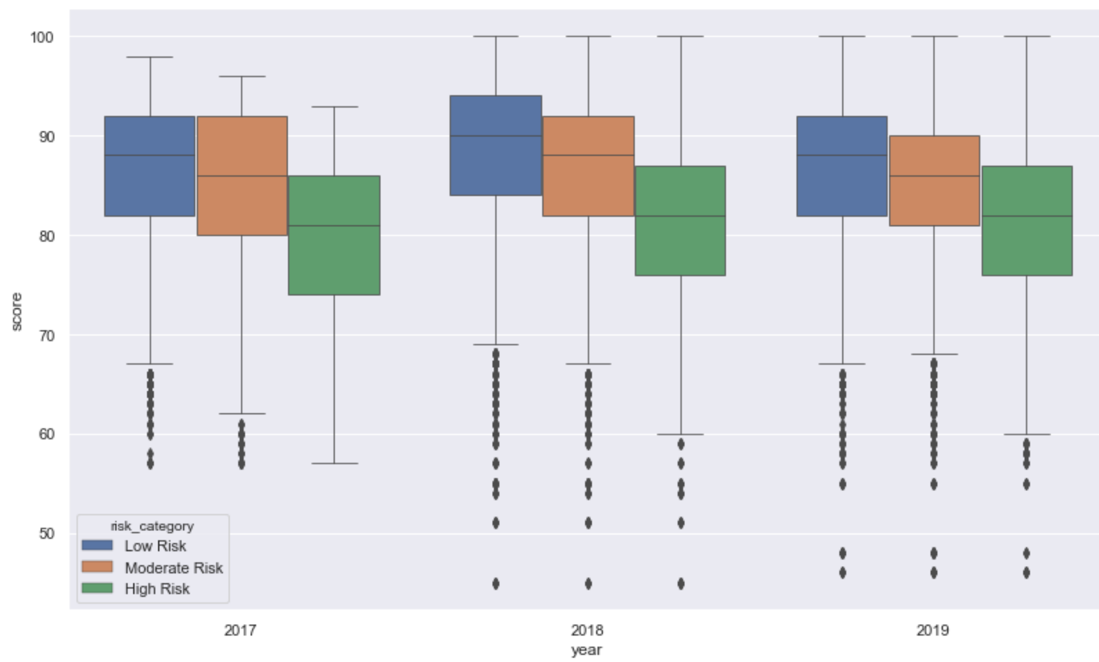
Hint: What does the slope represent?

The slope repesents no change in improvement. We see that scores are less likely to change if they are already high, but more likely to change with more variance if their first score was low, as shown in the upper right half of the graph.

0.1.3 Question 6d

To wrap up our analysis of the restaurant ratings over time, one final metric we will be looking at is the distribution of restaurant scores over time. Create a side-by-side boxplot that shows the distribution of these scores for each different risk category from 2017 to 2019. Use a figure size of at least 12 by 8.

The boxplot should look similar to the sample below. Make sure the boxes are in the correct order!

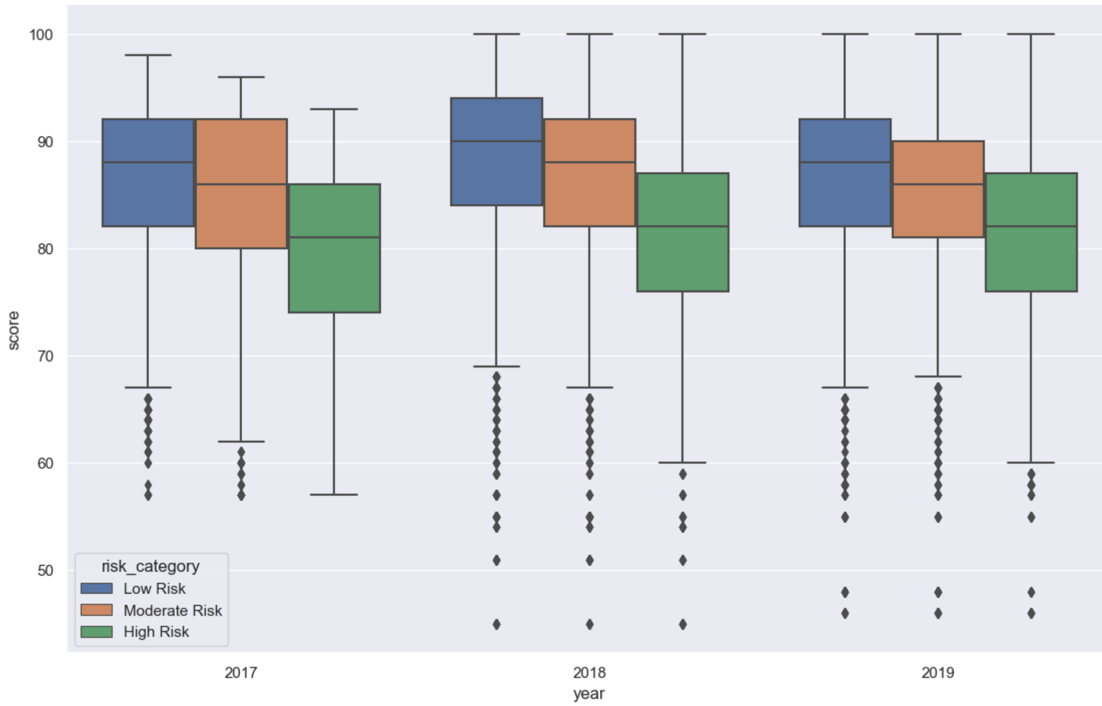


Hint: Use `sns.boxplot()`. Try taking a look at the first several parameters. The documentation is linked here!

Hint: Use `plt.figure()` to adjust the figure size of your plot.

```
In [78]: # Do not modify this line
         sns.set()
         plt.figure(figsize=(12, 8))
         risk = pd.merge(pd.merge(ins2vio, vio, on='vid'), ins, on='iid')
         risk = risk.loc[risk['year'] != 2016]
         sns.boxplot(x="year", y="score", data=risk, hue="risk_category", hue_order=['Low Risk', 'Modera
```

```
Out[78]: <AxesSubplot:xlabel='year', ylabel='score'>
```


score
100
90
80
70
60
50
year
2017
2018
2019
risk_category
Low Risk
Moderate Risk
High Risk

0.1.4 Grading

Since the question is more open ended, we will have a more relaxed rubric, classifying your answers into the following three categories:

- **Great** (4 points):
  - For a dataframe, a combination of pandas operations (such as groupby, pivot, merge) is used to answer a relevant question about the data. The text description provides a reasonable interpretation of the result.
  - For a visualization, the chart is well designed and the data computation is correct. The conclusion based on the visualization articulates a reasonable metric and correctly describes the relevant insight and answer to the question you are interested in.
- **Passing** (1-3 points):
  - For a dataframe, computation is flawed or very simple. The conclusion doesn't fully address the question, but reasonable progress has been made toward answering it.
  - For a visualization, a chart is produced but with some flaws such as bad encoding. The conclusion based on the visualization is incomplete but makes some sense.
- **Unsatisfactory** (0 points):
  - For a dataframe, no computation is performed, or the conclusion does not match what is computed at all.
  - For a visualization, no chart is created, or a chart with completely wrong results.

We will lean towards being generous with the grading. We might also either discuss in discussion or post on Piazza some exemplary analysis you have done (with your permission)!

You should have the following in your answers: * a question you want to explore about the data. * either of the following: * a few computed dataframes. * a few visualizations. * a few sentences summarizing what you found based on your analysis and how that answered your question (not too long please!)

Please limit the number of your computed dataframes and visualizations **you plan on showing** to no more than 5.

Please note that you will only receive support in OH and Piazza for Matplotlib and seaborn questions. However, you may use some other Python libraries to help you create you visualizations. If you do so, make sure it is compatible with the PDF export (e.g., Plotly does not create PDFs properly, which we need for Gradescope).

In [79]:
```
# YOUR QUESTION HERE (in a comment)
"""
Plot the  across San Francisco! Just like the photo shown in 2a.
"""
# YOUR DATA PROCESSING AND PLOTTING HERE

#Source: https://plotly.com/python/mapbox-county-choropleth/
```

```
sf_scores = pd.merge(ins, bus, on='bid')
sf_scores = sf_scores[sf_scores['latitude'] != -9999]
sf_scores = sf_scores[sf_scores['latitude'] != 0]

zline = sf_scores['score']
xline = sf_scores['latitude']
yline = sf_scores['longitude']

plt.scatter(xline, yline, c=zline, facecolors='none', edgecolors='b')
plt.colorbar()
plt.show()


# YOUR SUMMARY AND CONCLUSION HERE (in a comment)
"""
I first had to filter out data, where there were invalid long and lat, that would mess up the
plotted on to matplotlib. I tried to go further where I could underlay the map of San Francisc
near the pier and coast area. Maybe restaurants recieve popular views from there, and there is
affect location that heavily or vice versa. Maybe the bottom left area of SF could be said to
"""
```

```
Out[79]: "\nI first had to filter out data, where there were invalid long and lat, that would mess up th
```